Trait	Justification
Morphology (standard length, mass)	Morphological metrics, such as standard length and mass, can directly impact energetic requirements, thermoregulation, and life history traits, and therefore, reveal evolutionary trends that are likely to be beneficial under warmer conditions (Gardner et al., 2011; Verberk et al., 2021). Additionally, morphological metrics are not independent of fitness related traits, such as survivorship and fecundity, and can therefore suggest expected changes in reproductive success under different environmental conditions (Verberk et al., 2021). Combining standard length and mass metrics to assess body condition is hypothesized to estimate energetic reserves available to individuals that can be invested into expensive physiological processes (e.g., growth, reproduction) and can be considered an integrative measure of environmental quality, summarising the health and physiological state of an individual (Jakob et al., 1996). Furthermore, changes in morphology can further predict shifts in community level foodweb dynamics, as shifts in species biomass alter predator-prey interactions and behaviours (Audzijonyte et al., 2020; Cheung et al., 2013). However, previous results have found body condition to be an unreliable indicator of mortality under sustained increases in temperature, and therefore a poor indicator of survivorship (Rodgers et al., 2018).
Critical thermal maximum	Critical thermal maximum (CT_{max}) serves as a proxy for an organism's upper thermal tolerance. CT_{max} represent thermal conditions where organisms become incapacitated and can no longer escape stressful conditions, representing a hard boundary for species survivorship (Illing et al., 2020). CT_{max} has become a fundamental metric within fish ecology to understand the impacts of thermal stress on physiological performance and elucidate the impacts of warming temperatures, adaptive potential/capacity, and life-history strategies (Desforges et al., 2023). However, as outlined in a review by Desforges *et al*., (2023), researchers should be careful when interpreting results from CT_{max} experiments because results are sensitive to selected protocols (e.g., rate of thermal ramping, life stage, habituation periods; Desforges et al. 2023). It is therefore, not recommended to use CT_{max} as the only metric for an organism's upper thermal tolerance limits.
Oxygen uptake (intermittent-flow respirometry)/Critical thermal maximum/Critical temperature	To further explore physiological consequences associated with warming temperatures a resting oxygen uptake and critical thermal maximum experiments were performed simultaneously. Oxygen uptake serves as a proxy for metabolic rates (i.e., rates of ATP production by mitochondrial oxidative phosphorylation), and is used to understand plastic and evolutionary responses to environmental change (Killen et al., 2021). The dominant hypothesis for the physiological mechanism responsible for changing patterns in oxygen uptake under different thermal conditions if the oxygen and capacity limited thermal tolerance (OCLTT) hypothesis; as temperatures warm basal metabolism is progressively accelerated, leading to increased oxygen uptake from the environment (Pörtner et al., 2017; Pörtner & Farrell, 2008). However, there exists active debate around the universality of the OCLTT hypothesis (Lefevre et al., 2021; Schulte, 2015).

By measuring oxygen uptake simultaneously with CT_{max} measurements we were able to measure critical temperatures (T_c). T_c measurements refer to the temperature that separates permissive and stressful temperature domains (Ørsted et al., 2022). Physiologically T_c represents temperatures at organisms transition from temperatures effects on 'process supporting life' to those on 'processes causing injury and death' (Ørsted et al., 2022). Past T_c heat stress additively accumulates with increasing temperature intensity and duration, as a result of the rate of cellular damage outpacing the rate of cellular repair, subsequently causing the rate of heat failure to increase exponentially (Arnold et al., 2025; Ørsted et al., 2022). T_c therefore represents an more ecologically relevant metric than CT_{max} as it represents the temperature threshold that once past individuals can only withstand for a finite amount of time before it succumbs to heat stress (Arnold et al., 2025; Ørsted et al., 2022).

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